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Architecture for Distributed IoT Microgrids

Arquitectura Unified Namespace para Microrredes Distribuidas basadas en IoT

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ABSTRACT

In the context of the energy transition, electrical microgrids are crucial for integrating distributed renewable energy resources into modern power systems. Microgrid elements must operate within a common framework to ensure seamless dispatch and transition capabilities. For this reason, several IoT-based operational frameworks for generic distributed microgrid controllers have recently been proposed to support grid-connected and islanded modes. Commonly, many of these proposals leave the IoT communication architecture undefined, which is critical for guaranteeing reliable data exchange. This work introduces a Unified Namespace (UNS) architecture for an IoT-based distributed microgrids framework. The proposed approach establishes a UNS hierarchy and addresses key aspects, including message-exchange optimization, security, access control, and the implementation workflow.

Keywords: Architecture, Distributed Generation, Industrial Internet of Things, Microgrids, MQTT, Unified Namespace

RESUMEN

En el contexto de la transicion energetica, las microrredes electricas son cruciales para integrar los recursos energeticos renovables distribuidos en los sistemas electricos modernos. Los elementos que componen las microrredes deben funcionar dentro de un marco operativo comun para garantizar las capacidades de despacho y transicion. Por este motivo, recientemente se han propuesto diferentes marcos operativos basados en IoT para controladores genericos de microrredes distribuidas que admiten los modos conectado a la red e isla. Por lo general, estas propuestas no definen en detalle la arquitectura del sistema de comunicaciones IoT, un aspecto crítico para garantizar un intercambio de datos confiable. Este trabajo presenta una arquitectura de espacio de nombres Unificado (UNS) para un marco operativo de microrredes distribuidas basadas en IoT. El enfoque propuesto establece una jerarquía UNS y aborda aspectos clave como la optimizacion del intercambio de mensajes, la seguridad, el control de acceso y el flujo de implementación.

Palabras clave: Arquitectura, Generacion Distribuida, Internet Industrial de las Cosas, Microrredes, MQTT, Espacio de nombres Unificado

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Introduction

The global energy transition is underway, encompassing all the complex factors required to shift from fossil- based sources like coal, oil, and natural gas to renewable sources like solar, wind, and hydropower. This transition aims to reduce greenhouse gas emissions, improve energy efficiency, and mitigate global warming, ultimately contributing to a more sustainable future [1, 2]. This transformation process in how energy is generated is ongoing in many countries; however, several challenges remain in the successfully integration of distributed renewable energy resources into modern grids [3]. In this context, electrical microgrids (MGs)

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are critical to enabling this integration of energy sources, enhancing the resilience and efficiency of electrical grids [4]. In fact, it is expected that by 2030, about half a billion people will be powered by MGs [5].

A MG is a small-scale electrical system with clearly defined electrical boundaries that operates coordinately distributed generators and energy storage technologies to supply interconnected local loads [6]. This type of grid is an effective solution for integrating and managing distributed energy resources (DERs) within power systems due to its capability to operate in a decentralized and flexible manner [7]. An MG can operate in grid-connected mode, acting as a single controllable entity with respect to the grid, or in islanded mode [8, 9]. According to the IEEE Standard 2030.7 [10], the control system of an MG must consider at least two core functions: (a) **The dispatch function**, which dispatches DERs considering specified set-points; and

b) **The transition function**, which supervises transitions between grid-connected and islanded operation modes, coordinating the actions of DERs during this process. For these functions to operate properly, all MG components must work within a unified operation framework. This framework should ensure coordination, interoperability, scalability, and compliance with the regulatory operational requirements.

For this reason, standard operating frameworks for MGs have been proposed in the specialized literature. For example, [11] and [12] introduced generic MG controllers based on rule-based algorithms, while [13] proposed a self- healing MG controller. These proposals constitute a useful starting point, but their main limitation is the requirement for a central controller to coordinate and manage all DERs within the MG, leading to a single-point-of-failure risk.

Distributed frameworks have been proposed to overcome this limitation. For instance, [14] presents a multi-agent framework that employs a distributed economic dispatch algorithm in the grid-connected state, while a single DER handles frequency and voltage regulation in islanded mode. Similarly, [15] introduces a set of algorithms for grid- connected and islanded modes. These methods require designating a primary (leader) DER with a large power capacity that has access to critical signals and is followed by other DERs. This aspect may limit the implementation of such frameworks in applications where a large DER is not available.

To address these limitations, [16] presented an IoT-based framework for distributed generic MG controllers that supports MG operation during grid-connected and islanded modes, as well as during planned and unplanned transitions.

The proposal preserves the advantages of the distributed control paradigm by using generic components deployed on DER computational nodes and on the Point of Common Coupling (PCC) relay. This approach can easily adapt to new DER types, enabling plug-and-play capability and fully distributed operation.

The proposed framework focuses mainly on defining generic components for the computational nodes while leaving the architecture of the IoT communication system undefined and open to different alternatives. This is critical, as the IoT architecture directly impacts data-exchange reliability, which, in turn, affects service reliability for MG users. In this regard, the UNS architecture, a concept from industrial automation and digital transformation, emerges as a promising solution for the communication requirements of this IoT-based framework, offering operational advantages in terms of reliability, scalability, security, and efficiency.

This work proposes a UNS architecture for the IoT-based framework for generic distributed MG controllers presented in [16]. The proposed approach establishes a UNS hierarchy and addresses key aspects such as message exchange optimization, security, access control, and the implementation flow.

Compared with prior UNS deployments in industrial automation (largely conceptual or manufacturing-oriented), this work tailors UNS to the MG domain by: (i) outlining protocol-agnostic topic/payload guidelines (e.g., IEC 61850 and DNP3; compatible with SunSpec/CIM) that preserve electrical semantics; (ii) aligning the ISA-95 hierarchy with MG entities (operator, PCC relay, DER instances) to scope access and traffic; and (iii) specifying security and high-availability policies (mTLS/ACLs, broker clustering) constrained by MG operational needs. In contrast to the IoT framework in [16], which leaves the communications architecture open, we instantiate a concrete UNS-based communications layer for distributed MG control, including message optimization (topic filtering, edge aggregation, priority-aware QoS) and an implementation flow for onboarding and scaling DERs.

The rest of the paper is organized as follows. First, the main aspects of the IoT-based framework used as reference are described, including its computational nodes, message types, and current limitations. Next, the conceptual foundation of the UNS is introduced, outlining its key characteristics, operational advantages, and hierarchical structure. Subsequently, an adapted UNS architecture for distributed IoT MGs is presented, detailing the proposed hierarchy and relevant features of the deployment. Finally, the document presents the conclusions, highlights the main contributions, and outlines directions for future work.

IoT-Based Framework for Distributed Generic Microgrid Controllers

The proposal in [16] presents an innovative framework designed to enhance the management and operation of MGs by integrating IoT technologies. This framework supports the operation in both grid-connected and islanded modes, as well as during planned and unplanned transitions. It enables distributed control of MG components, real-time data collection, and localized decision-making, maintaining the benefits of a distributed control paradigm.

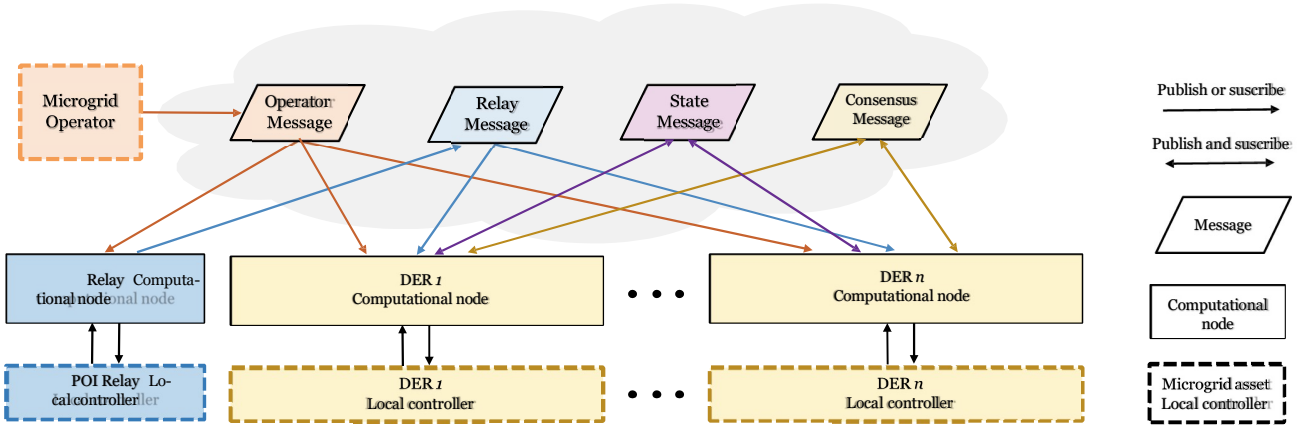


Figure 1. Adaptation of the IoT-based framework for distributed generic MG controllers proposed in [16].

The framework ensures seamless integration and coordi- nation by using generic components in the computational nodes of the DERs and the relay of the PCC (the node where the MG connects to the main utility grid). Furthermore, it adheres to standardized communication protocols, facilitat- ing plug-and-play capability and enabling fully distributed operation.

Fig. 1 presents an adaptation of the aforementioned framework. It comprises two types of computational nodes: the Point of Interconnection (POI) relay node and the DER node. Each node communicates with the local controllers of its respective MG assets and includes software components.

The **POI relay node** operates with a *Relay Control Component (RCC)* that collects relay measurements, evaluates whether the conditions for opening or closing the relay are met, and sends the corresponding commands. Additionally, this computational node includes a *Device I/O Component (DIOC)* that communicates with the end device's local controller using a protocol supported by that device (e.g., Modbus, DNP3, IEC 61850 GOOSE).

Similarly, the **DER nodes** are composed of a *State Machine Component (SMC)*, that determines the MG use case; a *Microgrid Computational Component (MCC)*, which determines the time step of the distributed algorithms; and their own *DIOC*. The software components described above are not included in Fig. 1 because they are not directly relevant to this work's proposal. For further details, please refer to the original paper.

The framework also defines the types of messages exchanged between computing nodes. At least four message types are specified:

- **Operator messages**: Published by the MG operator. The POI relay node and DER nodes subscribe to these messages to check for any requests regarding planned islanding or reconnection.
- **Relay messages**: Published by the POI relay node. The DER nodes subscribe to receive the POI relay status and islanding/reconnecting requests.
- **State message**: Each DER node publishes its local state using this message and subscribes to those published by other nodes. As a result, each DER node can receive the local states of all the DERs.
- **Consensus message**: Each DER node publishes consensus variables (e.g., per-unit active power, per-unit re active power, voltages, currents) and subscribes to the consensus messages of other DERs.

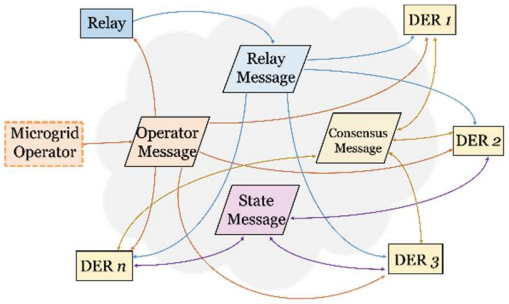


Figure 2. IoT message exchange using the framework for 4 DERs.

The proposed framework focuses primarily on defining the generic components of the computational nodes while leaving the architecture of the IoT communication system open to different alternatives. This is a critical aspect, as the IoT architecture directly impacts the security, stability, and efficiency of data exchange, thereby affecting MG users' service reliability. For example, as the number of DERs in the MG increases, IoT message exchange may become challenging due to the variety of message types and the need for both one-way and two-way communications. Fig. 2 illustrates this issue for a system with four DERs, noting that complexity grows significantly as more elements are integrated. Therefore, defining a scalable IoT architecture is essential to ensure seamless MG scalability, allowing new devices to be integrated without requiring major modifications to the existing infrastructure.

In this context, the UNS architecture presents a promising solution for meeting the communication requirements of this IoT-based framework. The following section explores the operational advantages of UNS for MG operations, including improved reliability, scalability, security, and efficiency.

Introduction to Unified Namespace

The UNS is a concept related to industrial automation and digital transformation that is still under development. As a result, most contributions in this field are found in blogs, webinars, and websites [17–21]. Some recent studies explore how UNS can address challenges related to fragmented data management within organizations [22], while others present practical cases of industrial implementation [23–25].

The UNS is a data architecture for industrial automation that organizes all entities, resources, and devices within a central hub or broker of organized data. This central hub serves as a single source of truth, providing a comprehensive framework that collects, organizes, and enables real- time access to data at all enterprise levels, facilitating communication between smart devices [17]. Essentially, the UNS represents both the business structure and all its real-time events, acting as the living repository of the organization's current state [26].

The UNS can interpret and contextualize data, facilitating seamless integration into various systems and platforms. This ensures that information generated by sensors and devices from diverse sources is standardized and uniformly understood. The UNS architecture is based on the publish- subscribe communication model, ensuring efficient data exchange via the MQTT protocol. This approach leverages the advantages of MQTT for machine-to-machine (M2M) communication and real-time management of large data volumes.

Adopting a UNS offers several operational advantages for organizations undergoing digital transformation within Industry 4.0. One of its key benefits is scalability, as the UNS eliminates the need for direct point-to-point integrations and enables the seamless incorporation of new IoT devices and sensors. Another critical advantage is data normalization, as the UNS standardizes information from heterogeneous sources and protocols. By harmonizing data structure, format, and representation, it ensures interoperability across diverse systems and applications within an integrated industrial architecture.

Additionally, UNS significantly reduces time-to-value by enabling real-time accessibility to data across the enterprise. Unlike conventional architectures that rely on multiple processing layers, the UNS provides immediate access to each data node, minimizing latency and improving decision- making agility.

The UNS also reinforces security through an advanced role-based access control system, ensuring that users can only access information relevant to their roles. Moreover, by facilitating a unified, real-time data flow, the UNS enhances agility. Eliminating data silos and ensuring continuous information availability allows teams to swiftly adapt to changes, make informed decisions, and implement adjustments without the delays imposed by conventional architectures.

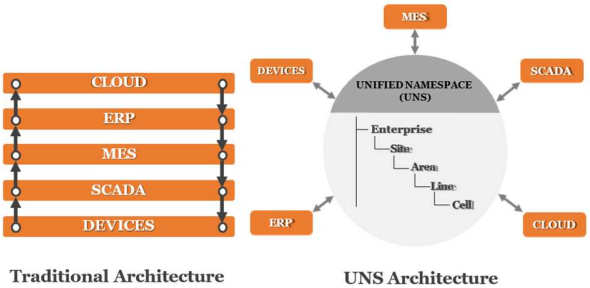


Figure 3. Traditional and UNS Architecture

UNS hierarchy

The levels of the international ISA-95 standard (also known as ANSI/ISA-95 or IEC 62264) define a hierarchical model for manufacturing systems, from the physical process to enterprise planning, as illustrated to the left of Fig. 3, [27]. Level 1 includes field devices such as sensors and actuators; Level 2 covers control systems like PLCs and SCADA; Level 3 represents the Manufacturing Execution System (MES), responsible for managing and monitoring production operations; Level 4 refers to Enterprise Resource Planning (ERP) systems, which handle business planning, logistics, and resource management; and finally, Level 5, extending beyond traditional ISA-95, typically represents cloud-based systems, enabling broader data integration, analytics, and remote services.

The UNS is fully compatible with the ISA-95 structure, adapting each of its levels to a radial model, as shown to the right of Fig. 3. Maintaining a consistent topic structure is essential for defining a predictable namespace.

In protocols such as MQTT, the structured nature of topics enables the implementation of a UNS hierarchy that streamlines the distribution and access of data within an organization. Typically, the UNS hierarchy levels are organized as follows:

Enterprise > Site > Area > Line > Cell

As this hierarchy can be adapted to the requirements of the ISA-95 standard for manufacturing systems, it can also be easily adapted to other applications and system types, such as the MG controllers. By leveraging IoT technologies, MG controllers can optimize performance, improve real- time decision-making, and facilitate remote supervision. The next section explores this application in greater detail, highlighting its benefits and potential use cases.

Proposed approach

Based on the IoT framework described above, this section proposes an adapted UNS architecture to address communication and scalability challenges in distributed MGs. The approach integrates the hierarchical structure of the UNS with the specific requirements for publishing and subscribing to different types of messages, ensuring reliable, secure, and efficient data exchange between the computational nodes associated with the DERs and the POI relay.

UNS Architecture

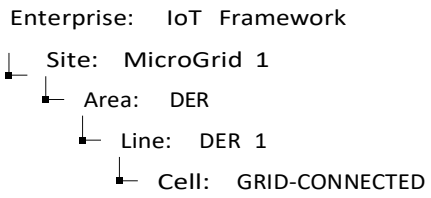
A UNS architecture is proposed to integrate the IoT MG controller framework into a UNS-based hierarchical namespace, as shown in Fig. 4. ISA-95 levels are mapped to MG roles and assets to scope data and control, while protocol-aware normalization and control are exposed through a single, enforceable namespace that preserves electrical semantics and keeps applications protocol-agnostic.

- Enterprise Level: This upper level covers the complete IoT framework.
- Site Level: This level encompasses each MG and facilitates message exchange between MGs if such functionality is integrated into a higher control layer of the framework.
- Area Level: This level encompasses the different types of MG assets, such as the MG operator, the relay, and

the DERs, which serve as the primary operational areas.

- Line Level: This level refers to the individual elements of the MG assets, such as the MG operator, the POI relay, and all DERs connected to the MG.
- Cell Level: This level represents the messages exchanged between the different computational nodes.

This hierarchical structure applies the ISA-95 hierarchy to MG operations by mapping levels to roles and assets: Enterprise (global project), Site (MG), Area (role domain: Operator, PCC relay, DER), Line (asset instance, e.g., DER_1), and Cell (message domain: Measurements, Events, Commands). Implemented using MQTT topics, it standardizes data organization, enforces least-privilege access via topic-scoped ACLs, and scales efficiently as assets are added. For example, a GRID-CONNECTED state message from a DER could be structured as follows:



The proper hierarchy of messages in the UNS architecture ensures that namespace levels align with the application's scope and message type. This proposal recommends assigning all messages to the cell level. However, general messages, such as Operator Messages, could be assigned to higher levels as Area or Line to reduce redundancy and simplify access. This hierarchical structure, a key feature of the UNS, requires a thorough analysis of information flows, component relationships, and operational requirements. By doing so, the namespace structure can be tailored to the system's specific needs, balancing flexibility with resource optimization.

The DIOC in each node handles protocol translation (e.g., Modbus to MQTT), while the UNS broker manages unified data flow. This implementation enhances plug-and- play capabilities and distributed operation, addressing the limitations identified in the original proposal.

Field data and controls are normalized by the DIOC into a canonical topic/payload model so that upstream applications never embed protocol-specific logic. Under the UNS hierarchy, steady telemetry is published in measurement namespaces, and time-critical changes are published in event namespaces; commands use a dedicated command path with acknowledgements. Payloads carry a value along with normalized quality and timestamp information, preserving semantics while providing uniform consumption and access control across roles.

For IEC 61850, Logical Node/Data Object/Data Attribute paths map to measurement topics, GOOSE/SMV messages are published as high-priority events, and controls are sent on command topics with explicit acknowledgements. For DNP3, Object Group/Variation/Index paths map the same way to measurement topics, and control functions are issued on command topics with result codes in the acknowledgement.

Beyond those protocols, topics include CIM tags so that assets and points are easy to discover across enterprise systems. IEEE 1547 grid-support functions and states are carried as standard measurements, configurations, events, and commands. SunSpec models map to deterministic topics, with units and scales provided in metadata, and writable points issued via the command path with acknowledgements. In short, CIM supplies the vocabulary, IEEE 1547 defines the behavioral rules, and IEC 61850/DNP3/SunSpec provide the field interfaces|all unified within a single canonical topic/payload model so that upstream applications remain protocol-agnostic.

Optimization of Message Exchange

The communication scheme for distributed MGs faces the challenge of managing a growing volume of messages. As shown in Figure 2, the diversity of message types and the need for both unidirectional and bidirectional communication add complexity, particularly as more DERs connect to the MG. To ensure efficient operation and system scalability, the UNS architecture incorporates three key mechanisms:

- **Topic Filtering**: DERs and the PCC relay subscribe only to relevant topics, reducing unnecessary data traffic. For example, for Operator Message updates DERs subscribe to

  Enterprise > Site > Area: Operator Message

  and for Relay Message updates DERs subscribe to

  Enterprise > Site > Area > Line > Cell: Relay.

- **Data Aggregation**: To reduce network congestion, DERs implement local data aggregation mechanisms before publishing to the UNS. Instead of transmitting high-frequency raw measurements, each DER processes its raw data and publishes it at periodic intervals. For instance, rather than broadcasting every consensus update, a DER may report the final agreed value at the end of each control cycle. This strategy minimizes redundant transmissions, reduces bandwidth usage, and ensures that only high-value information reaches other subscribers.
- **Quality of Service (QoS)**: MQTT QoS levels can be assigned based on message priority. High-priority messages (e.g., relay commands) can use QoS 2 to ensure delivery, while low-priority data (e.g., periodic state updates) can use QoS 1 or 0.

Security and Access Control

The operation of MGs demands a robust security framework that protects data integrity and system availability. The UNS architecture addresses these requirements through a layered approach that combines:

- **Role-Based Access Control (RBAC):** Allows to define permissions for the MG operator, DERs, and the relay nodes. For example, only the MG operator can publish Operator Messages, while DERs can only publish their own state and consensus messages.

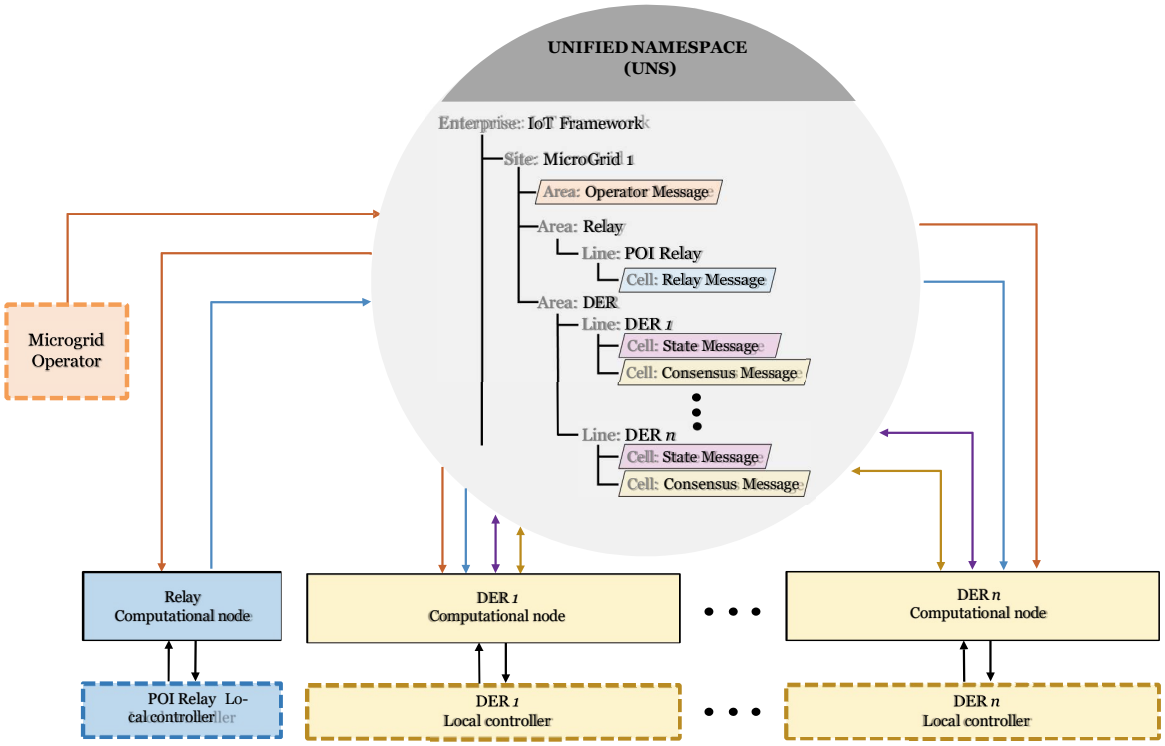


Figure 4. Proposed UNS architecture for the IoT-based framework for distributed generic MG controllers.

- **Encryption and Protection Against Failures/Attacks**: All MQTT traffic is encrypted in transit with TLS, enforcing a minimum protocol version and modern cipher suites; client-broker handshakes reject weak/legacy options. For control messages, the system adds anti-replay metadata (timestamps) and, where required, payload signatures to assure integrity beyond transport. To withstand failures or hostile traffic, the broker applies defense-in-depth (limits per client/topic rate-limiting, publish quotas, and in-flight window caps-along) with message TTLs and retained- message quotas to prevent backlog amplification.
- **Authentication and Credentials**: Each node authenticates with unique credentials, ensuring only trusted devices and roles can connect to the broker. Authorization is enforced with topic-level access control lists (ACLs) across the UNS hierarchy: Operators may publish only under Operator Message; each DER may publish only under its own State and Consensus Message paths and may subscribe to Relay and Operator Message; Relays publish their status and accept commands under Relay Message. This enforces RBAC with least-privilege, and all other actions are denied by default.
- **High Availability and Redundancy**: To eliminate single points of failure, the UNS is deployed as a redundant, multi-broker cluster with shared subscriptions and horizontal load balancing. Broker nodes are distributed across availability zones and fronted by a resilient endpoint (DNS/Anycast/virtual IP) so clients reconnect transparently. Session state and retained messages are persisted/replicated to enable takeover when needed, while inter- site bridging replicates critical topics for cross-site continuity. Client agents are provisioned with multiple bootstrap endpoints, publish LWT (Last Will and Testament) health signals, and apply exponential back-off with in-flight caps to prevent reconnect storms.

Implementation Flow

The deployment of the proposed UNS architecture follows a systematic process designed to minimize disruptions to existing operations. Deployment is structured into four main phases:

- **Namespace Initialization:** The UNS broker is configured with a standardized hierarchical topic structure and security policies, ensuring compatibility across heterogeneous DERs and diverse protocols via MQTT.
- **Node Registration:** DERs and the PCC relay register with the broker, subscribing to relevant topics in a distributed manner, eliminating single points of failure associated with centralized controllers.
- **Real-Time Operation:** DERs publish state and consensus messages at defined intervals, the PCC relay monitors relay status and publishes commands, and the MG operator publishes area-level directives, enabling rapid responses to disturbances or operator commands.
- **Scalability:** New DERs integrate easily by registering with the broker and subscribing only to the topics relevant to their role-no changes to existing communication nodes are required. As the number of DERs grows, message volume increases primarily with the number of active publishers and their update rates. To keep performance predictable, the UNS applies (i) topic partitioning aligned with the Enterprise/Site/Area/Line/Cell hierarchy to bound subscription scopes; (ii) edge aggregation at the DIOC to coalesce fast-changing points and publish

at the control-cycle cadence; and (iii) priority- aware QoS (e.g., QoS 2 for interlocking/commands and QoS 0{1 for periodic telemetry). Together, these mechanisms prevent pairwise fan-out and keep per-DER traffic growth approximately linear under nominal conditions.

Conclusions

The UNS architecture presented in this work provides a scalable and efficient communication framework for distributed MGs, addressing key challenges in interoperability, message management, and security through its MQTT-based hierarchical structure. By organizing messages according to their operational scope (from DER-specific data to global operator commands), network traffic is optimized, and redundancy is reduced, while the robust security scheme ensures system integrity. The proposed implementation demonstrates how UNS can adapt to the dynamic needs of MGs, facilitating both the integration of new devices and coordinated decision- making.

This work aims to support both the academic and industrial communities in adopting UNS, highlighting its potential applications in the electrical sector. Future work will focus on the practical aspects of this proposal, presenting the results of an implementation that is currently under development.

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Author contributions

All authors made significant contributions to the research and provided critical comments. Juliam A. Díaz contributed to the Conceptualization, Methodology, Validation and Writing. Joaquín Delgado contributed to the Conceptualization and Methodology. Iván Hernández contributed to the Conceptualization, Methodology and Writing Review. Natalia Duarte contributed to the Conceptualization, Methodology and Writing Review. Juan
M. Rey contributed to the Conceptualization, Methodology and Writing Review.

References

[1] N. Zhao and F. You, "Can renewable generation, energy storage and energy efficient technologies enable carbon neutral energy transition?" *Applied Energy*, vol. 279, p. 115889, 2020, https://doi.org/10.1016/j.apenergy.2020.115889.

[2] Q. Hassan, S. Algburi, A. Z. Sameen, T. J. Al-Musawi, A. K. Al-Jiboory, H. M. Salman, B. M. Ali, and M. Jaszczur, "A comprehensive review of international renewable energy growth," *Energy and Built Environment*, 2024, https://doi.org/10.1016/j.enbenv.2023.12.002.

[3] M. Khalid, "Smart grids and renewable energy systems: Perspectives and grid integration challenges," *Energy Strategy Reviews*, vol. 51, p. 101299, 2024, https://doi.org/10.1016/j.esr.2024.101299.

[4] M. S. Alam, F. S. Al-Ismail, A. Salem, and M. A. Abido, "High-level penetration of renewable energy sources into grid utility: Challenges and solutions," *IEEE Access*, vol. 8, pp. 190277–190299, 2020, https://doi.org/10.1109/ACCESS.2020.3031481.

[5] E. ESMAP, "Mini grids for half a billion people: Market outlook and handbook for decision makers," *World Bank*, 2019. [Online]. Available: https://www.esmap.org/mini_grids_for_half_a_billion_people_the_report

[6] M. Farrokhabadi, C. A. Can˜izares, J. W. Simpson-Porco, E. Nasr, L. Fan, P. A. Mendoza-Araya, R. Tonkoski, U. Tamrakar, N. Hatziargyriou, D. Lagos, R. W. Wies, M. Paolone, M. Liserre, L. Meegahapola, M. Kabalan, A. H. Hajimiragha, D. Peralta, M. A. Elizondo, K. P. Schneider, F. K. Tuffner, and J. Reilly, "Microgrid stability definitions, analysis, and examples," *IEEE Trans. on Power Systems*, vol. 35, no. 1, pp. 13–29, 2020, https://doi.org/10.1109/TPWRS.2019.2925703.

[7] J. M. Rey, G. A. Vera, P. Acevedo-Rueda, J. Solano, M. A. Mantilla, J. Llanos, and D. Saéz, "A review of microgrids in latin america: Laboratories and test systems," *IEEE Latin America Transactions*, vol. 20, no. 6, pp. 1000–1011, 2022,https://doi.org/10.1109/TLA.2022.9757743.

[8] D. T. Ton and M. A. Smith, "The U.S. depart- ment of energy's microgrid initiative," *The Electric- ity Journal*, vol. 25, no. 8, pp. 84–94, 2012, https://doi.org/10.1016/j.tej.2012.09.013.

[9] J. M. Rey, P. P. Vergara, J. Solano, and G. Ordo´ n˜ez, "Design and optimal sizing of microgrids," *Microgrids Design and Implementation*, pp. 337–367, 2019. [Online]. Available: https://link.springer.com/chapter/10. 1007/978-3-319-98687-6_13

[10] "IEEE standard for the specification of microgrid controllers," *IEEE Std 2030.7-2017*, pp. 1–43, 2018, https://doi.org/10.1109/IEEESTD.2018.8340204.

[11] G. Razeghi, F. Gu, R. Neal, and S. Samuelsen, "A generic microgrid controller: Concept, testing, and insights," *Applied Energy*, vol. 229, pp. 660–671, 2018, https://doi.org/10.1016/j.apenergy.2018.08.014.

[12] C. Sun, G. Joos, S. Q. Ali, J. N. Paquin, C. M. Rangel, F. A. Jajeh, I. Novickij, and F. Bouffard, "Design and real-time implementation of a centralized microgrid control system with rule-based dispatch and seamless transition function," *IEEE Transactions on Industry Applications*, vol. 56, no. 3, pp. 3168–3177,
2020, https://doi.org/10.1109/TIA.2020.2979790.

[13] J. Lee, G. Razeghi, and S. Samuelsen, "Generic microgrid controller with self-healing capabilities," *Applied Energy*, vol. 308, p. 118301, 2022, https://doi.org/10.1016/j.apenergy.2021.118301.

[14] M. H. Cintuglu, T. Youssef, and O. A. Mohammed, "Development and application of a real-time testbed for multiagent system interoperability: A case study on hierarchical microgrid control," *IEEE Transactions on Smart Grid*, vol. 9, no. 3, pp. 1759–1768, 2018, https://doi.org/10.1109/TSG.2016.2599265.

[15] X. Hou, Y. Sun, J. Lu, X. Zhang, L. H. Koh, M. Su, and J. M. Guerrero, "Distributed hierarchical control of ac microgrid operating in grid-connected, islanded and their transition modes," *IEEE Access*, vol. 6, pp. 77 388–77 401, 2018, https://doi.org/10.1109/ACCESS.2018.2882678.

[16] H. Tu, H. Yu, Y. Du, S. Eisele, X. Lu, G. Karsai, and S. Lukic, "An IoT-based framework for distributed generic microgrid controllers," *IEEE Transactions on Control Systems Technology*, vol. 32, no. 5, pp. 1692–1705, 2024, https://doi.org/10.1109/TCST.2024.3378989.

[17] K. Manditereza, "Implementing unified namespace (UNS) with MQTT Sparkplug," 2022.

[Online]. Available: https://www.hivemq.com/article/ implementing-unified-namespace-uns-mqtt-sparkplug/

[18] "What is unified namespace (UNS) and why does it matter?" 2022. [Online]. Available: https://www.hivemq.com/article/ what-is-unified-namespace-uns-iiot-industry-40/

[19] "How does a unified namespace (UNS) work?" 2022. [Online]. Available: https://www.hivemq.com/article/how-does-unified-namespace-uns-work-iiot-industry-40/

[20] C. Systems, "Mapping unified namespaces to domain namespaces," 2024. [Online]. Available: https://corsosystems.com/posts/mapping-unified-namespaces-to-domain-namespaces

[21] W. Reynolds, "Digital factory mastermind: Methodology and architecture to deploy industry 4.0 solutions for manufacturers," 2024. [Online]. Available: https://www.iiot.university/digital-factory-mastermind

[22] Á. Péter and S. Werner, "The impact of Unified Namespace in Industry 4.0," 2024, Department of Automatic Control, Lund University. Advisor: Prof. Charlotta Johnsson. [Online]. Available: https://lup.lub. lu.se/student-papers/search/publication/9174552

[23] S. Domíguez-Cid, D. F. Larios, F. J. Molina, J. Luque, A. Carrasco, and C. Leo´n, "SSOT- Based laboratory on IIoT for Industry 4.0 archi- tectures," *16th Congreso de Tecnologia, Aprendizaje y Ensenanza de la Electronica, TAEE 2024*, 2024, https://doi.org/10.1109/TAEE59541.2024.10604981.

[24] P. Koprov, S. Gadhwala, A. Walimbe, X. Fang, and B. Starly, "Systems and methods for authenticating manufacturing machines through an unobservable fingerprinting system," *Manufacturing Letters*, vol. 35, p. 1009 { 1018, 2023, https://doi.org/10.1016/j.mfglet.2023.08.051.

[25] J. Maule, T. Kutzler, T. Leich, D. Schafer, and C. Richter, "Unified namespace and asset administration shell: A winning combination for digital production," *2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA)*, pp. 1–7, 2024, https://doi.org/10.1109/ETFA61755.2024.10710821.

[26] W. Reynolds, "4.0 Solutions: Free 1-hour unified namespace (UNS) training webinar," 2024. [Online]. Available: https://www.youtube.com/watch?v=ogRy8_ut_AA

[27] K. Mandittereza, "Designing your uns semantic information hierarchy," 2023. [Online]. Available: https://www.hivemq.com/blog/designing-your-uns-semantic-information-hierarchy/